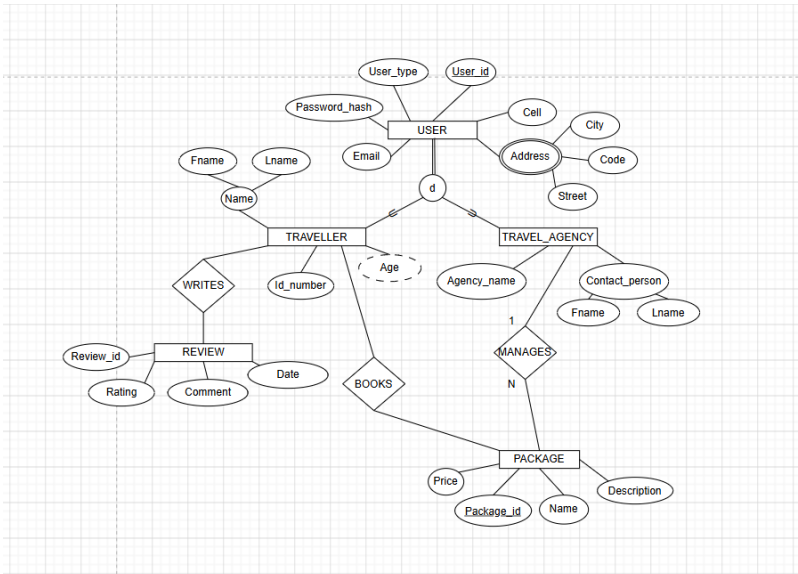
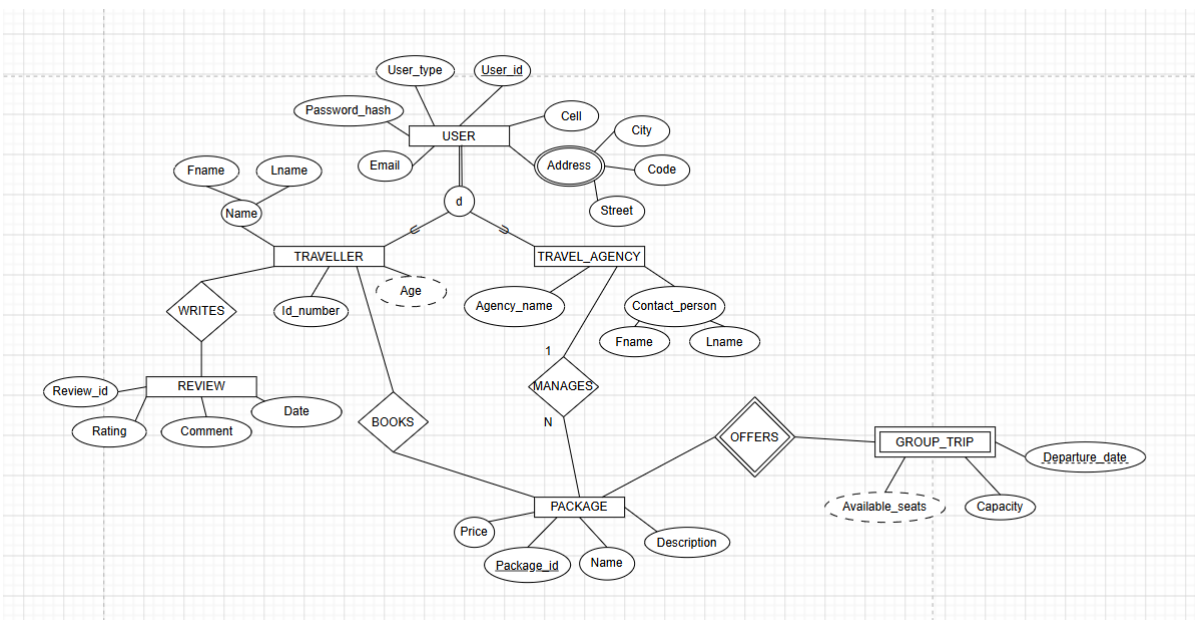


EER Diagram

TASK 2:

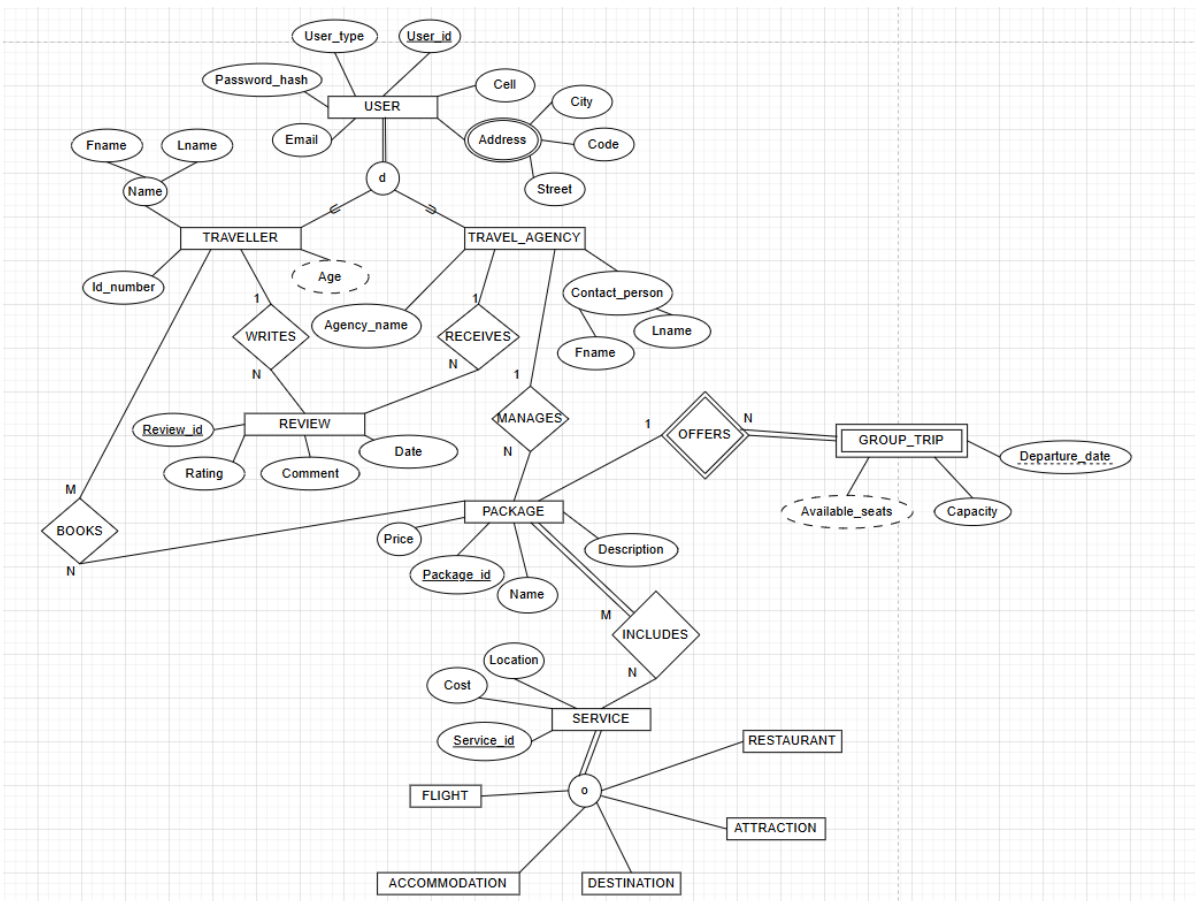


First iteration.



Added ability to make group trips based on departure date

I made User_id the primary key because an email or contact number may change over time.



Added SERVICE because PACKAGES includes many different services.

Then on the final EERD, I gave attributes to each SERVICE subtype.

To ensure agencies are trustworthy, I added a AGENCY_LICENSE entity to easily check up on their credibility.

Added PROMO_CODE because a specific traveller adds a specific promo_code to a specific package

Added REVIEW_TARGET because TRAVEL_AGENCY and PACKAGE acts as a review target. (People write reviews about them)

Multivalued, Composite Attributes

Address as a multivalued & composite attribute because people or agencies might operate at multiple locations simultaneously, and each location requires a city, postal code, and street address.

Total Specialization & Disjoint Constraint

I applied a disjoint (d) constraint with a total specialization rule (double line) to the USER supertype. I assumed that every account in the system is strictly required to be **EITHER a Traveller OR a Travel Agency**, and they **CANNOT** share roles.

Weak Entities

GROUP_TRIP is a weak entity that is dependent on PACKAGE.

Overlapping Constraint

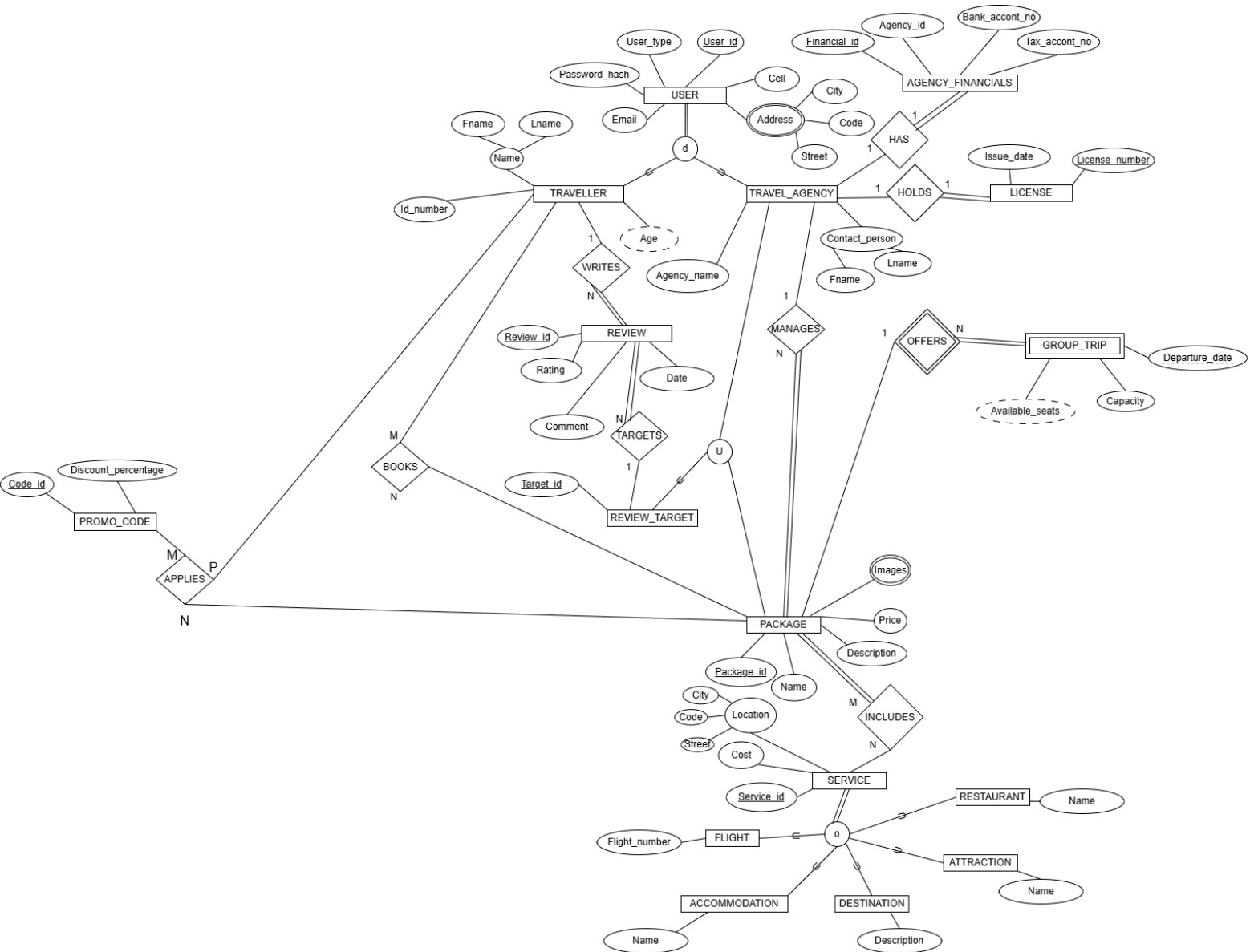
I applied an overlapping (o) constraint to the SERVICE super type. I assumed that a single venue could fulfil multiple roles simultaneously (an establishment acting as both a RESTAURANT and an ATTRACTION).

Derived Attributes

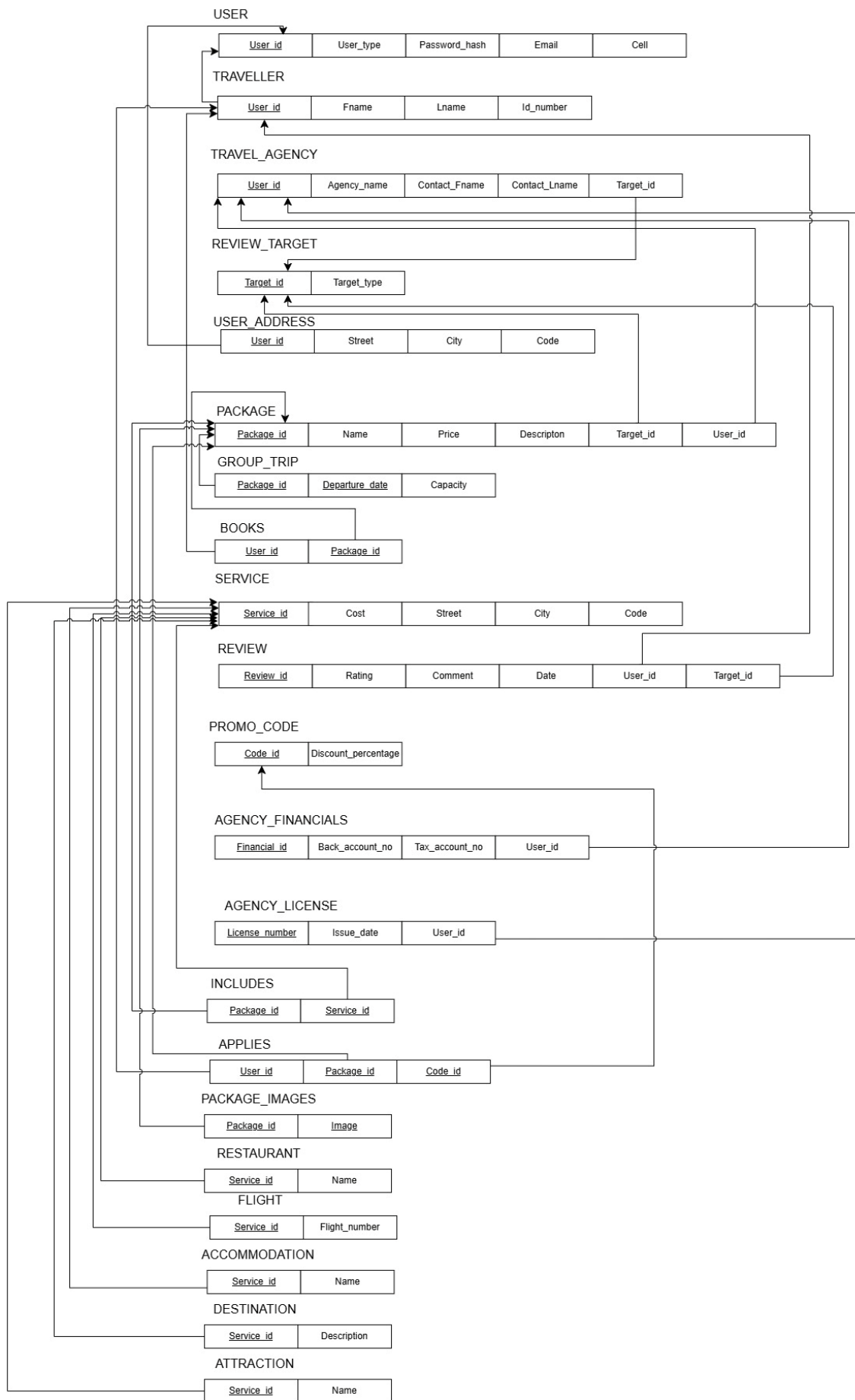
I assumed that a traveller's Age and a group trip's Available_seats can be calculated dynamically at runtime (Capacity - Booked Tickets and from ID numbers). I show them as derived attributes (dashed ovals).

Many-to-Many Relationships

I included an M:N INCLUDES relationship between PACKAGE and SERVICE. I assumed that a single curated travel package bundles multiple services, and a single catalogued service can be reused across many different packages.



TASK 3:



ER-to-Relational Mapping Algorithm

- **Step 1: Mapping of regular (strong) entity types**

- For each regular entity type, E, in the ER schema
- Create a relation R
- Add simple attributes of E to R
- Include only the simple component attributes of composite attributes
- Choose one of the key attributes of E as the primary key for R. If the attribute is a composite, then use all the simple attributes of the composite for the primary key of R

USER(User_id, User_type, Password_hash, Email, Cell)

PACKAGE(Package_id, Name, Price, Description)

SERVICE(Service_id, Name, Location, Cost)

REVIEW(Review_id, Rating, Comment, Date)

- **Step 2: Mapping of weak entity types**

- For each weak entity type, W, in the ER schema with owner entity E
- Create a relation R
- Add the simple attributes of W to R
- Include as foreign key attributes of R, the primary key of the relations that correspond to the owner entity type
- The primary key of R is a combination of the primary key(s) of the owner(s) and the partial key of W, if any
- If there is a weak entity type **E1**, whose owner is also a weak entity type, **E2** then **E1** should be mapped before **E2** to determine the primary keys first.

primary key is a combination of the parent's primary key and the weak entity's partial key

GROUP_TRIP(Package_id, Departure_date, Capacity)

- **Step 3: Mapping of binary 1:1 relationships**

LICENSE(License_number, Issue_date, User_id)

AGENCY_FINANCIALS(Financial_id, Bank_AccountNo, Tax_AccountNo, User_id)

- **Step 4: Mapping of binary 1:N relationships (FK on N side)**

These are handled by including Foreign Keys (FK) in the "N" side of the relationship.

- **Step 5: Mapping of binary M:N relationships**

Primary key is both primary keys of participating entities

BOOKS(User_id, Package_id)

INCLUDES(Package_id, Service_id)

- **Step 6: Mapping of multivalued attributes**

A separate table is created for each multivalued attribute. (Note: Flattening is used.)

USER_ADDRESS(User_id, Street, City, Code)

PACKAGE_IMAGES(Package_id, Image_URL)

- **Step 7: Mapping of N-ary relationships**

Primary key combination of all participating primary keys

APPLIES(User_id, Package_id, Code_id)

- **Step 8: Mapping specialisation and generalization**

TRAVELLER(User_id, Fname, Lname, Id_number)

TRAVEL_AGENCY(User_id, Agency_name, Contact_Fname, Contact_Lname)

FLIGHT(Service_id, Flight_number)

RESTAURANT(Service_id, Name)

ATTRACTION(Service_id, Name)

ACCOMMODATION(Service_id, Name)

DESTINATION(Service_id, Description)

- **Step 9: Mapping unions**

REVIEW_TARGET(Target_id, Target_type)

TASK 4:

